

(a) SHOCKSPEC at firing τ_j

one record per firing, $j \leq 2$

shock_family
direction
onset_window

names $\hat{k}_j$
— dates $E_{j\hat{k}_j}$

stream, magnitude, persistence,
duration, evidence, signature:
not read by the hedge

*abstain, no parse, or outside $\mathcal{H}$:
no $\hat{k}_j$ is named*

the model cannot write

✗ an order ✗ a threshold
✗ a likelihood ✗ code

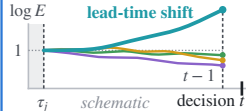
(b) Prior over fixed $\mathcal{H}$

mass $a_j w_{jk}$ to each $k \in \mathcal{H}$

■ demand up 1/16
■ demand down 1/16
■ lead-time shift 13/16
■ upward pulse 1/16
■ $\lambda/|\mathcal{H}|$ each
■ $+(1 - \lambda)$ to the named $\hat{k}_j$

$a_j = \alpha 2^{-j}$, $A = \sum_j a_j \leq 3\alpha/4$
 $w_{jk} = \lambda/|\mathcal{H}| + (1 - \lambda) \mathbb{1}\{k = \hat{k}_j\}$
registered hedge $\lambda = 1/4$

no $\hat{k}_j$: every $w_{jk} = \lambda/|\mathcal{H}|$

(c) Certify: evidence to posterior mass

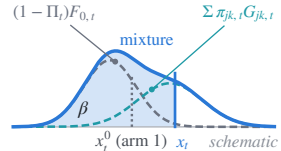
one e-process per (j, k) ,
 $E_{jk, \tau_j} = 1$, data after τ_j ;
anytime valid under $\mathbb{P}_0$;
the model's onset
window dates $E_{j\hat{k}_j}$

$$\pi_{jk, t} = \frac{a_j w_{jk} E_{jk, t-1}}{1 - A_t + N_{t-1}}$$

$$\begin{aligned} N_t &= \sum_{j, k} a_j w_{jk} E_{jk, t} \\ A_t &= \sum_{j: \tau_j < t} \sum_k a_j w_{jk} \\ \Pi_t &= \sum_{j, k} \pi_{jk, t} \end{aligned}$$

size from the data, within each k

■ ■ ■ multipliers, starts and durations; the level
learned from 5 pseudo-observations
offset $\Delta \in \{1, 2, 3\}$ and start, from receipts
 $\Rightarrow$ law $G_{jk, t}$, not the model's magnitude bin

(d) Hedge: the order q_t 

$\beta = p/(p + h)$ quantile of the mixture:
 $x_t = \inf \{x : (1 - \Pi_t)F_{0, t}(x + P_t) + \sum_{j, k} \pi_{jk, t} G_{jk, t}(x + P_t) \geq \beta\}$
 $q_t = \min(\max(0, \lceil x_t - I_t \rceil), C_t, \hat{C}_t)$

$\Pi_t = 0$: q_t is arm 1's order, bit for bit

(e) Guarantees

Exposure under $\mathbb{P}_0$: known null, registered arrival law
 $\mathbb{E}_0[\Pi_\sigma] \leq A$ at every stopping time σ of $(\mathcal{F}_{t-1})$,
for any model output, adversarial included

Odds regret
 $c_{F_0}(x_t) - c_{F_0}(x_t^0) \leq \Pi_t V_t / (1 - \Pi_t)$
one-period newsvendor surrogate

Evidence, not validity both against $\lambda = 1$
✓ named: $+\log 3.25$ nats
✗ wrong or missing family: $-\log 4$ nats